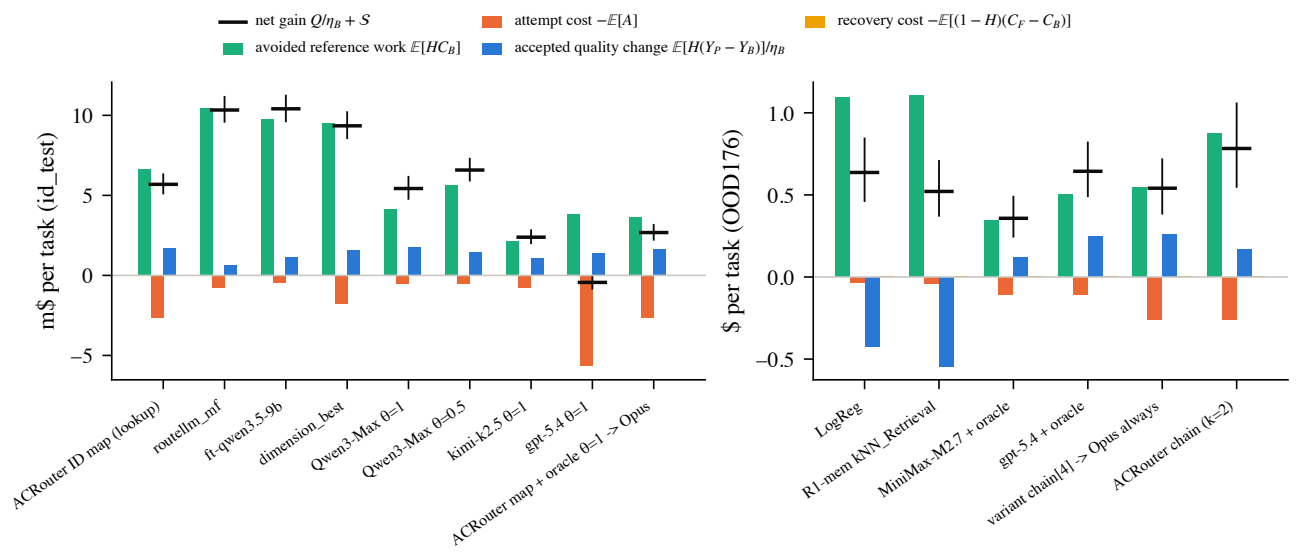
net gain $Q/\eta_B + S$
attempt cost $-\mathbb{E}[A]$
recovery cost $-\mathbb{E}[(1-H)(C_F - C_B)]$
avoided reference work $\mathbb{E}[HC_B]$
accepted quality change $\mathbb{E}[H(Y_P - Y_B)]/\eta_B$
m$ per task (id_test)
10
5
0
−5
ACRouter ID map (lookup)
routellm_mf
ft-qwen3.5-9b
dimension_best
Qwen3-Max θ=1
Qwen3-Max θ=0.5
kimi-k2.5 θ=1
gpt-5.4 θ=1
ACRouter map + oracle θ=1 -> Opus
$ per task (OOD176)
1.0
0.5
0.0
−0.5
LogReg
R1-mem kNN_Retrieval
MiniMax-M2.7 + oracle
gpt-5.4 + oracle
variant chain[4] -> Opus always
ACRouter chain (k=2)